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1. Overall Summary

This paper addresses the challenge of optimizing cache storage and content fetching in Cognitive Radio Networks (CRNs) where content popularity is unpredictable and primary user band occupation is dynamic. Existing DRL methods often treat the system as a single agent or homogeneous agents, overlooking the autonomous nature of users and cache servers, and are limited to discrete action spaces. The authors propose a novel heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach. In this model, users and Secondary Base Stations (SBSs) are treated as distinct types of agents that learn to cooperate and compete. Users aim to minimize delivery delay, while SBSs optimize cache storage for maximum efficiency and revenue. The MADDPG framework allows SBSs to optimize the cached fraction of each content in a continuous space, offering greater flexibility than discrete eviction/retention. With centralized training and decentralized execution, agents can optimize actions individually based on partial observations, reducing communication overhead. Numerical simulations demonstrate that this heterogeneous MADDPG significantly reduces user delivery delay and enhances SBS cache efficiency, outperforming single-agent or homogeneous DRL approaches (like DDPG, PPO, TD3) by nearly 40% in total system reward, proving its effectiveness in balancing mixed cooperative-competitive objectives.

2. Main Idea

The central concept is to propose a heterogeneous multi-agent deep deterministic policy gradient (MADDPG) approach for cooperative caching in Cognitive Radio Networks (CRNs). This approach treats both users and Secondary Base Stations (SBSs) as two distinct types of agents that learn to cooperate and compete to reduce user delivery delay and enhance SBS cache efficiency simultaneously. Key innovations include optimizing cached content fractions in a continuous space and leveraging centralized training with decentralized execution.

3. Problem Being Solved

This paper aims to solve several weaknesses and research gaps in previous cooperative caching systems for Cognitive Radio Networks (CRNs). Specifically, it addresses the difficulty of dynamically optimizing cache storage and content fetching strategies due to unpredictable content requirements and primary user band occupation. Previous DRL approaches either model multi-cache as homogeneous agents or the entire system as a single global agent, failing to account for the autonomous and often conflicting objectives of individual users and cache servers. Additionally, existing DRL methods like DQN often operate in discrete action spaces, limiting the flexibility of cache management. Centralized learning approaches also incur redundant communication overhead. The paper tackles how to maximize storage efficiency for cache nodes, how users can minimize delivery delay by selecting the best cache node, and how to design an effective learning algorithm for heterogeneous agents in a co-existing cooperation and competition environment.

4. Assumptions

- The slot duration T_0 is long enough that all content requests of users can be served within a time slot.
- SBSs are geographically closer to users than PBS, leading to a faster transmission rate from SBSs ($r_{n,k} > r_{0,k}$).
- The overall contents consist of F files, and the file size s_f is independent identically distributed (i.i.d) of Pareto distribution.
- Contents request vector $q_k(t)$ follows Zipf distribution.
- Each file can be divided into fragments using a rateless code like Raptor codes.
- For user k , it can only retrieve one fragment of file f from SBS n with the size of $A_{n,f}$, and retrieve the rest fraction $(1-A_{n,f})$ from PBS, rather than employing multiple SBSs to cooperatively relay the fragments.

5. Limitations

- The proposed approach was tested on a specific simulation configuration with a fixed number of SBSs and users, and specific geographical distribution.
- The model assumes that a user can only retrieve one fragment of a file from a single SBS and the rest from the PBS, not from multiple SBSs cooperatively relaying fragments, which might limit the potential for more complex cooperative strategies.

6. Results and Conclusion

The numerical simulations demonstrate the effectiveness and superiority of the proposed heterogeneous MADDPG approach. User and SBS rewards consistently converge, confirming the algorithm's stability. The caching strategy adapts to SBS location and replacement costs, with farther SBSs prioritizing hotter content and higher replacement costs leading to more focused caching of popular content. Compared to homogeneous MADDPG variants, the heterogeneous MADDPG significantly improves both user rewards (lower delivery delay) and SBS rewards (higher cache efficiency). Furthermore, it outperforms other state-of-the-art DRL algorithms like DDPG, PPO, and TD3, achieving a nearly 40% improvement in total system reward. The final conclusion is that the heterogeneous MADDPG effectively balances the mixed cooperative-competitive relationship between users and SBSs, leading to simultaneous optimization of content storage and delivery delay, and achieving a better global reward in CRNs.

7. Real World Impact

This research has practical real-world applications in optimizing wireless network performance. By enabling distributed cooperative caching in Cognitive Radio Networks, it can significantly decrease network traffic loads, reduce content delivery delay for users, and free up valuable spectrum resources. This is particularly relevant for scenarios like edge AI, smart cities, and mobile broadband where efficient content delivery and spectrum utilization are critical. The proposed approach can lead to more responsive and efficient wireless communication systems, improving user experience and network resource management in dynamic and unpredictable environments.

8. Graphs and Figures Analysis

Fig. 1: System model

This figure illustrates the system architecture. It shows a Primary Base Station (PBS) connected to multiple Secondary Base Stations (SBSs), which in turn serve multiple users. The diagram highlights two access paths: users primarily access content from SBSs (solid line) and secondarily from PBS if not available at SBSs (dashed line). It also depicts the multi-agent learning framework, showing K user agents and N SBS agents, each with an 'Actor' (π) and a 'Critic' (Q) network. The 'Actor' takes state (S) and action (A) to produce a reward (R), while the 'Critic' evaluates the action. The training process involves global state (S_All) and global action (A_All) for the critics, while execution is decentralized.

Fig. 2: Locations and rewards of heterogeneous agents

This figure presents three sub-figures related to the simulation setup and initial results. Figure 2(a) is a scatter plot showing the geographical locations of the PBS, five SBSs (SBS1-SBS5), and three users (User1-User3) in a 2D coordinate system, visualizing the network topology. Figure 2(b) is a line graph showing the reward for each of the three users over 300,000 training episodes. All user rewards converge to a stable negative value, indicating that the learning process stabilizes and users achieve a consistent minimum delivery delay. Figure 2(c) is a line graph showing the reward for each of the five SBSs over 300,000 training episodes. All SBS rewards converge to stable positive values, demonstrating that SBSs successfully optimize their cache efficiency and revenue. SBS2 and SBS4 show slightly lower rewards, consistent with their farther location from users.

Fig. 3: Contents popularity vs. cached fraction

This figure shows two sub-figures illustrating the impact of SBS location and replacement cost on caching strategy. Figure 3(a) is a line graph showing the average cached fraction (λ) as a function of normalized content popularity for each of the five SBSs. It shows that more popular contents are cached by a larger fraction. SBS2 and SBS4, being farther from users, exhibit slightly different caching strategies, tending to cache the hottest contents more aggressively to compensate for their location disadvantage. The shaded area indicates the storage capacity limit. Figure 3(b) is a line graph showing the average cached fraction (λ) as a function of normalized content popularity for SBS1, under different replacement cost factors ($\beta = 0.5, 0.8, 1.0, 1.2$). As β increases (higher replacement cost), the curves shift from up-concave to down-convex, indicating that SBSs prioritize caching more popular contents when the cost of replacement is higher or cache space is scarce.

Fig. 4: Comparison with homogeneous MADDPG

This figure presents two sub-figures comparing the overall reward of users and SBSs across different MADDPG scenarios. Figure 4(a) is a line graph showing the average reward of all users over 300,000 training episodes for three scenarios: "Heter-MADDPG (users and SBSs)", "Homo-MADDPG (SBSs Only)", and "Homo-MADDPG (users Only)". The blue line (proposed heterogeneous MADDPG) achieves the highest and most stable negative reward (lowest delay), outperforming the homogeneous MADDPG variants. Figure 4(b) is a line graph showing the average reward of all SBSs over 300,000 training episodes for the same three scenarios. The blue line (proposed heterogeneous MADDPG) achieves the highest and most stable positive reward, indicating superior cache efficiency and revenue compared to the homogeneous MADDPG variants.

Fig. 5: Performance comparison with other DRL

This figure is a line graph comparing the global reward (sum of user and SBS rewards) of the proposed heterogeneous MADDPG with other DRL algorithms (DDPG, PPO, and TD3) over 300,000 training episodes. The proposed MADDPG (blue line) consistently achieves the highest global reward and converges faster and to a higher value than DDPG, PPO, and TD3, demonstrating its superior performance in balancing user delivery delay and SBS cache efficiency in a mixed cooperative-competitive environment. The total reward is improved by nearly 40% compared to homogeneous learning approaches.